Introduction to Artificial Intelligence

What is Artificial Intelligence?

Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. These processes include learning, reasoning, and self-correction. AI systems can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.

Types of AI

There are three main types of AI: Narrow AI, General AI, and Super AI. Narrow AI, also called Weak AI, is designed to perform a specific task, such as facial recognition or internet searches. General AI, or Strong AI, can perform any intellectual task that a human can do. Super AI surpasses human intelligence in all respects.

Machine Learning

Machine Learning is a subset of AI that gives systems the ability to automatically learn and improve from experience without being explicitly programmed. It focuses on developing computer programs that can access data and use it to learn for themselves. The process begins with observations or data to look for patterns in data and make better decisions in the future.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with many layers (hence deep) to analyze various factors of data. It is inspired by the structure and function of the human brain. Deep learning is responsible for significant advancements in computer vision, natural language processing, and speech recognition.

Natural Language Processing

Natural Language Processing (NLP) is a branch of AI that helps computers understand, interpret, and manipulate human language. NLP draws from many disciplines including computer science and computational linguistics. Applications include chatbots, language translation, sentiment analysis, and text summarization.

RAG - Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) is an AI technique that combines information retrieval with text generation. Instead of relying solely on knowledge stored in model parameters, RAG retrieves relevant documents from an external knowledge base and uses them to generate more accurate and up-to-date responses. This approach reduces hallucinations and allows the model to cite sources.

Vector Databases

Vector databases store data as high-dimensional vectors, which are mathematical representations of features or attributes. They are essential for AI applications because they enable fast similarity search across large datasets. ChromaDB, Pinecone, and Weaviate are popular vector databases used in RAG systems to store and retrieve document embeddings.

Large Language Models

Large Language Models (LLMs) are AI models trained on vast amounts of text data. They can generate human-like text, answer questions, summarize documents, translate languages, and perform many other language tasks. Examples include GPT-4, Claude, and Gemini. These models form the foundation of modern AI assistants and chatbots.

AI Ethics

AI ethics involves ensuring that AI systems are developed and used in ways that are fair, transparent, and beneficial to society. Key concerns include bias in AI systems, privacy, accountability, and the impact of AI on employment. Organizations worldwide are developing guidelines and regulations to ensure responsible AI development.

Future of AI

The future of AI holds enormous potential. AI is expected to transform industries including healthcare, education, finance, and transportation. Advances in areas like multimodal AI, which can process text, images, and audio together, and AI agents that can autonomously complete complex tasks, will shape the next decade of technology.